

METEROLOGY: WHEATHER PARAMETER, CROP WHEATHER ADVISORY

Last 5 Year (2019-2023) Previous Year Question

- 1. The outermost layer of the atmosphere. In this layer, atoms and molecules escape into space and satellites orbit the orbit. The name of layer is _____. (2023)**
 - a. exosphere
 - b. stratosphere
 - c. Troposphere
 - d. Mesosphere

- 2. Which of the following Weather factor that affects water measures in plants? (2019)**
 - a. Temperature
 - b. Humidity
 - c. Wind velocity
 - d. Rainfall

- 3. Dry region of Tamil Nadu receives highest rainfall from which monsoon (2022)**
 - a. South west monsoon
 - b. North east monsoon
 - c. Remain dry in both monsoonal wind
 - d. Both a & b

- 4. What is the percentage of carbon dioxide in the atmosphere? (2021)**
 - a. 78.9 %
 - b. 0.033%
 - c. 0.33%
 - d. 0.0003%

- 5. What is the percentage of nitrogen in the atmosphere? (2019)**

- a. 70.8 %
- b. 71%
- c. 78.088 %
- d. 21%

6. According to India Meteorological Department (IMD), a rainy day has been defined as a day with rainfall of

- a. More than 1.5 mm rainfall
- b. Less than 2.5 mm rainfall
- c. More than 2.5 mm rainfall
- d. None of the above

Answers

1. A 2.D 3.B 4.B 5.C 6.C

METEROLOGY: WHEATHER PARAMETER, CROP WHEATHER ADVISORY

Crop growth and development are primarily governed by environmental conditions i.e. climate & soil. **The success and failure of crops is intimately related to the weather during the crop periods.** A sound knowledge of the climatic factors and its interactions with crop is essential for successful agriculture.

Some Important Terminologies

1. Weather

It is a state or condition of atmosphere at a given place and at a given time.

- ✓ It is the **daily variation** or condition of lower layers of the atmosphere.
- ✓ Weather pertains for a **very smaller region** like a village, city for a smaller duration of time period.

Example- rainy day, cold night, hot noon or sunny day, a foggy day etc.

2. Climate

Climate is a generalized weather or **summation of weather condition** over a given region during a comparatively longer period.

- ✓ Climate is a weather condition **related to larger areas** like zone, state, country etc. during a month, season or a year.

Example- tropical climate, monsoonal climate etc.

3. Meteorology

Meteorology is defined as a branch of physics that deals with the **physical process in the atmosphere that produce weather**.

4. Agriculture meteorology

It is the branch of applied meteorology which deals **with the response of crops to the physical environment**.

- ✓ It is the study of physical process of **the atmosphere that produces weather in relation to agricultural production, and also deals with agroclimatology, instrumentation and weather forecasting**.

5. Climatology

It is the science that deals with **the factors which determine and control the distribution of climate over the earth's surface**.

6. Agroclimatology

It deals with the relationship of **climatic regimes and agricultural production**.

- It includes instrumentation and weather forecasting.

ATMOSPHERE

Atmosphere is the **gaseous envelope** that surrounds the earth surface it is tasteless, colour less, and odorless mixture of gasses which **extends up to 1600 km height above the earth surface**.

The 99% mass of the total atmosphere is available **within 40 km** from the earth surface.

Composition Of The Atmosphere

A sample of pure air consist of **78% of nitrogen, 21 % of oxygen, 0.9% of argon, and 0.033%** of carbon dioxide. Very minor amount of neon, helium, methane, krypton, hydrogen, ozone, etc. comprise the remaining fractional percentage of atmosphere.

➤ **Lower atmosphere contains water vapor from 0 to 4 % by volume.**

Table 1 – Composition Of Gases

<u>GASES</u>	<u>VOLUME</u>
Nitrogen	78.088 (78%)
Oxygen	20.948 (21%)
Argon	0.930 (0.9%)
Carbon dioxide	0.033 (0.3%)
Other gases	Traces

Structure Of Atmosphere

Based on vertical temperature difference, atmosphere is notionally divided into four major layers **with no clear or distinct boundaries** due to mixing of gases upwards and downwards across the boundaries.

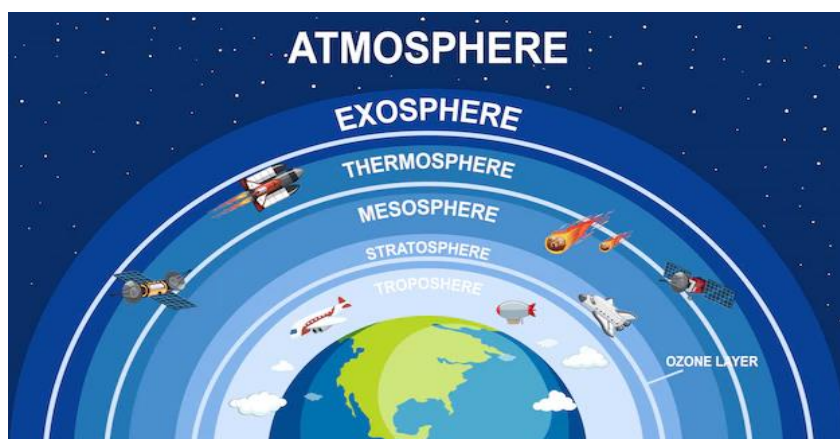


Image (i)- Layers of Atmosphere

1. Troposphere

- ✓ Lowermost layer_extended up to height of 8 to 18 km from earth surface.
- ✓ Thicker at equator then on poles.
- ✓ **All weather phenomenon like clouds, fog, dew, mist rain etc. occurs in this layer.**

- ✓ Temperature decreases with the height and at the boundary that **is tropopause temp is -60°C .**

2. Stratosphere

- ✓ Lies above tropopause.
- ✓ Above **18 km height and extends up to 50 km** height.
- ✓ It is dust free, cloudless warmest layer.
- ✓ **Temperature increases with increase in height.**
- ✓ Presence of **ozone layer** in this region prevent the earth surface and organism from UV rays coming from sun.

3. Mesosphere

- ✓ Lies above the stratosphere.
- ✓ Mesosphere is **separated from stratosphere by a thin layer** called stratopause.
- ✓ Decrease in temperature **with increase in altitude (height).**
- ✓ It extends from **50 km to 80 km.**

4. Thermosphere

- ✓ Separated by mesopause from mesosphere.
- ✓ Outermost layer that merges with the space.
- ✓ There **is a presence of ions** in its lower layer hence called as **a ionosphere.**
- ✓ **Long range communication satellites** are placed in this layer due to presence of charged ions.

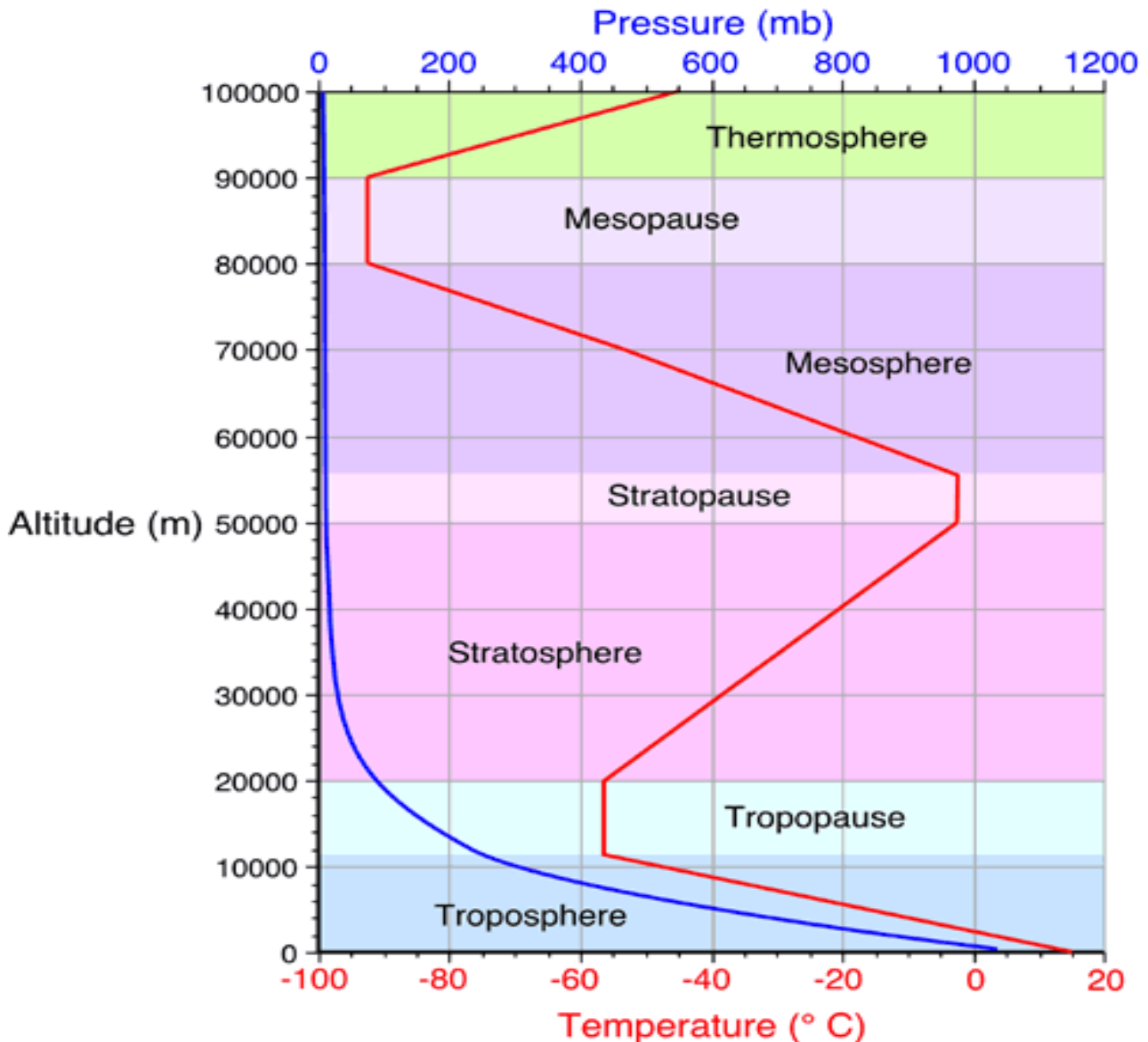


Image (ii)- layers of atmosphere

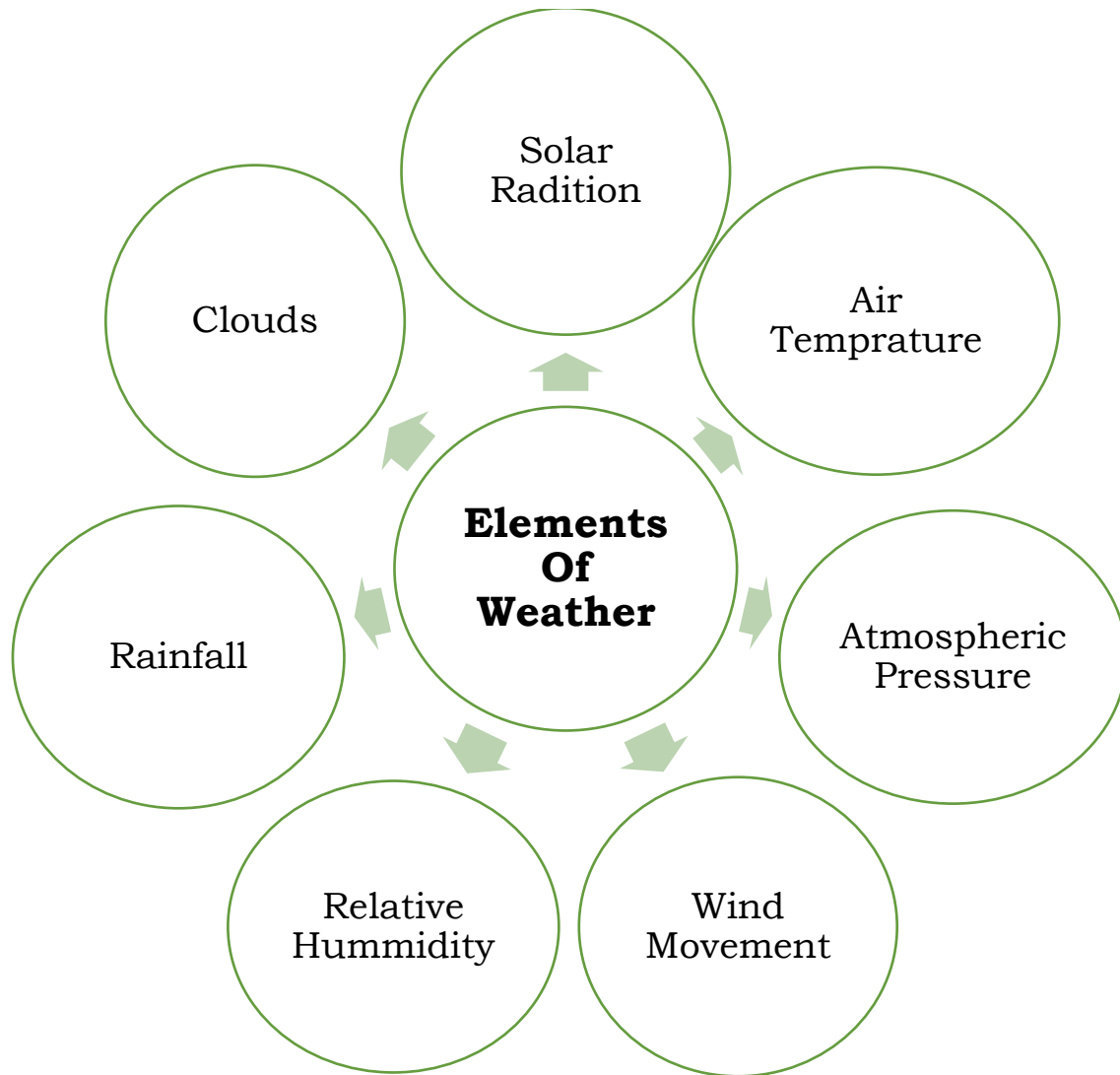
WEATHER ELEMENTS

When it is desired to give the state of **weather at any particular time** and at a particular place, the following elements are considered.

1. Solar radiation

- ✓ The incoming energy from the sun is called as a solar radiation.
- ✓ A constant amount of energy is emitted by the sun that is **1.94cal/cm²/min. this value is also called as solar constant.**
- ✓ Solar radiation is received in the form of electromagnetic waves.
- ✓ Instrument used to measure total incoming radiation is called as pyranometer.

- ✓ Photosynthetically active radiation **PAR -is light of wavelengths 400-700 nm** and is the portion of the light spectrum utilised by plants for photosynthesis.
- ✓ PAR above the crop surface is measured by tube solarimeters.



Flow Chart 1. Elements Of Weather

2. Temperature

- ✓ The **degree of hotness & coldness of** a substance, determined by the extent of its molecular activity and is measured by a thermometer.
- ✓ Heat flow between any substance can takes place by **three processes known as conduction, convection, radiation.**
- ✓ **Conduction**- transfer of heat by molecular activity. Ex heat transfer between solid bodies
- ✓ **Convection**-transfer of heat between liquids and gases by the motion of fluids.
- ✓ **Radiation** -transfer of heat by electromagnetic waves.

3. Wind-

- ✓ Wind is **air in horizontal motion** which travels from **a high-pressure area to a low-pressure area**
- ✓ Speed of the wind is called as wind velocity which is measured by **anemometer**.
- ✓ Direction of the wind is measured by **wind pane**.

4. Humidity

- ✓ Humidity refers to the **amount of water vapor present** in the air
- a. Absolute humidity**- It is the actual quantity of the water vapor by weight present in the given volume of the air.

$$\text{Absolute humidity} = \frac{\text{Weight of water vapour}}{\text{Volume of the air}}$$

- b. Specific humidity** is the weight of water vapour per unit weight of the air including the water vapour.
- ✓ It is expressed as grams **of water vapour per kilogram of air**.

$$\text{Specific humidity} = \frac{\text{Weight of the water vapour}}{\text{Weight of air including water vapour}}$$

- c. Relative humidity** is the ration between the **amount of water vapour present in the air and the amount of water vapour required for** saturation at a particular temperature and pressure. It is expressed as percentage or ratio.

$$\text{Relative humidity} = \frac{\text{Water vapour present in the air}}{\text{Water vapour required for saturation}} \times 100$$

Relative humidity can be measured using **wet bulb or dry bulb thermometer**.

Atmospheric pressure-

It is defined as a pressure **exerted by the air column of air with a cross-sectional area** of a given unit extending from the earth surface to the upper boundary of the atmosphere.

It is measured with the help of aneroid barometer and barograph.

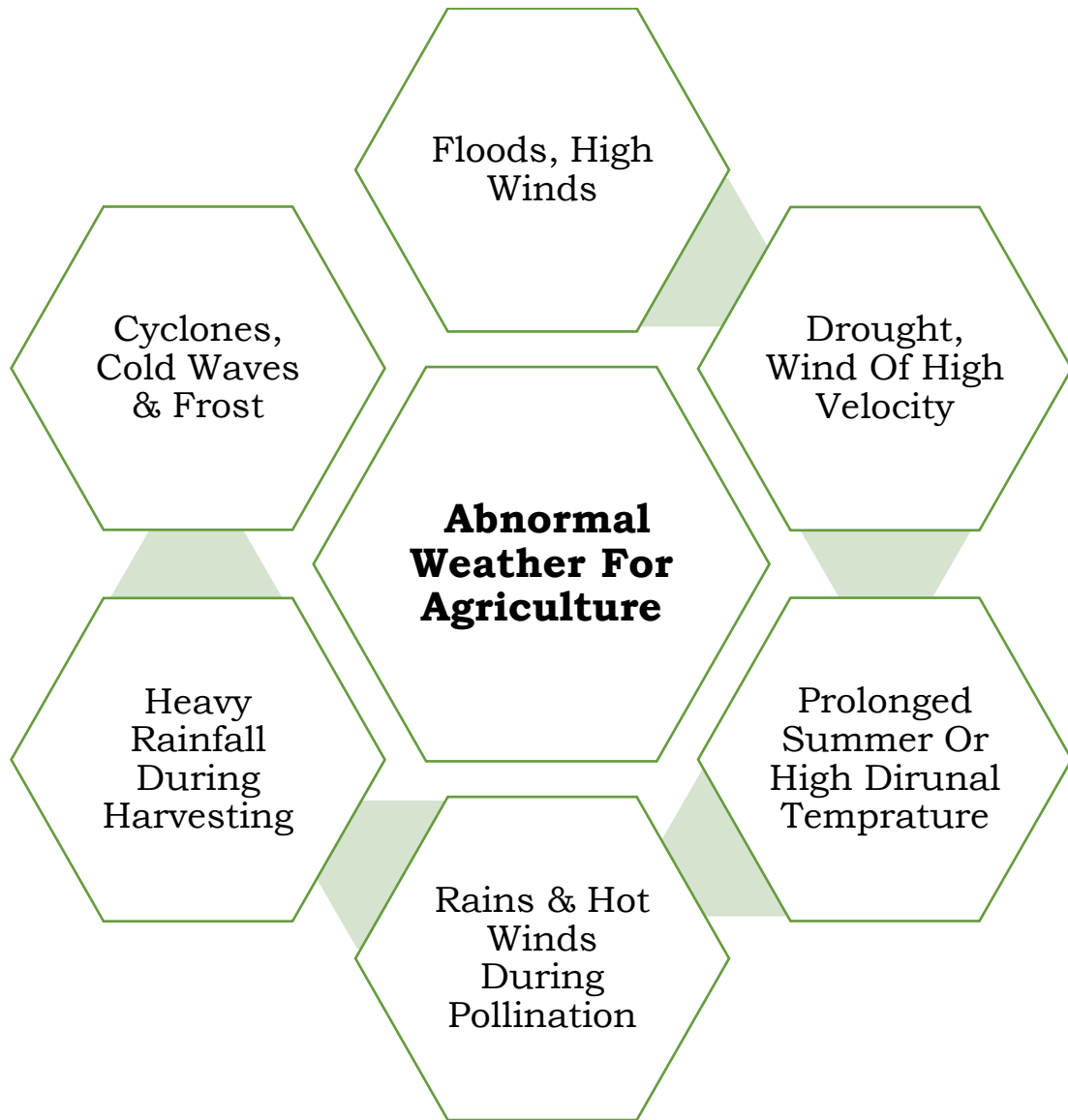
Rainfall

- ✓ Rainfall is the form of precipitation in the **liquid drops larger than 0.5 mm in diameter**. Falling on the earth surface
- ✓ Diameter of rain drop **varies from 0.5 to 4 mm**.
- ✓ India **receives 70% of rainfall** from the south west monsoon.
- ✓ **Dry spell**- rainless day of **more than 10 days for light soil conditions and 15 days for heavy soil conditions**.
- ✓ **Rainy day** – when **more than 2.5mm** rainfall is received on a particular day
- ✓ **Sowing rain**- when farmers take up sowing of crops in the rainfall is **more than 20 mm during** July and this rainfall is called as sowing rain.
- ✓ **Effective rainfall** -it means **the useful or utilizable rainfall for the agriculture** which provide the required moisture for the growing crop after reducing the amount of runoff, evaporation and wastage is called as effective rainfall.
- ✓ **EL Nino**-is used widely to refer abnormal warming of the ocean resulting weaker south west monsoonal rainfall in the Indian peninsula.
- ✓ **La Nina**- it is the **opposite EL Nino which** relates with the abnormally colder eastern Pacific Ocean.

Clouds- As the air **mass goes up it expands due to low pressure and** cools. If the cooling proceeds up to saturation water vapour condenses and cloud formation takes place.

ABNORMAL WEATHER FOR AGRICULTURE

Weather is not always normal. **Some abrupt or sudden changes that takes place in the weather which are called as weather abnormalities**. These weather abnormalities can cause heavy damage to crops, livestock and human life



Flow Chart 2- Abnormal Weather for Agriculture

WEATHER FORECASTING

“Weather forecast is an expression of probability **of a particular future state of the atmospheric system in a given point or territory**”.

- ✓ **Indian meteorological department** was established in 1875 with headquarters at Pune.
- ✓ **Agriculture meteorological division was started in 1932** for conducting research on crop weather relationships.
- ✓ There are **five regional forecasting** centers in India that are in **Chennai, Mumbai, Nagpur, Delhi & Calcutta**.

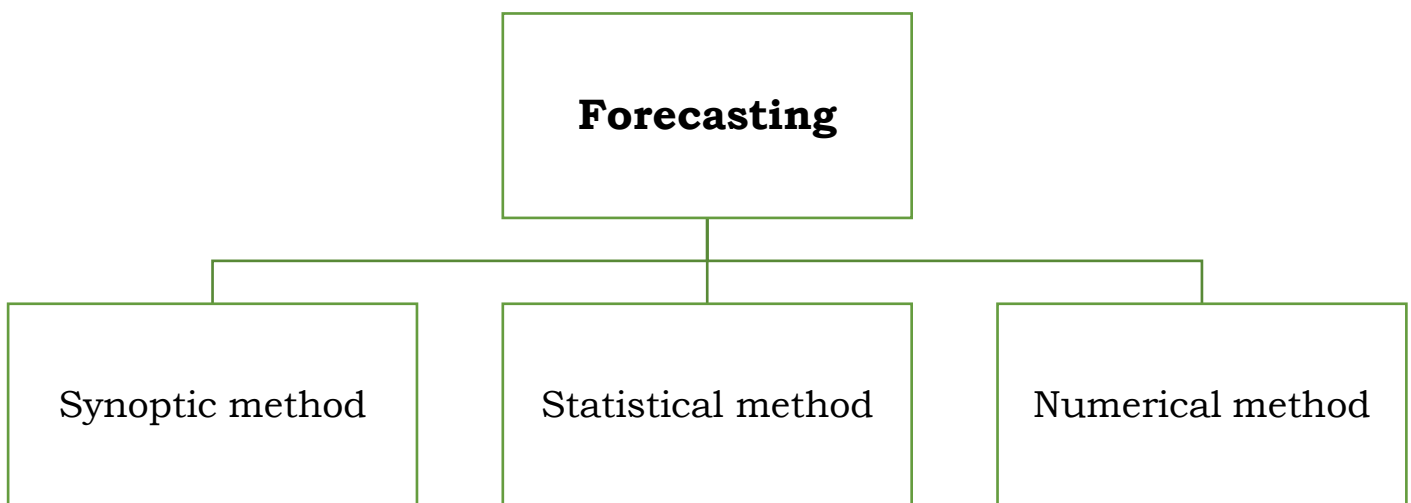
These regional centers prepare weather forecast called as weather bulletins which indicates the **probable date of onset of monsoon, intensity, duration, breaks in rainfall and adverse weather phenomenon**. Which is telecasted by radio or television in the regional languages.

Essentials for weather forecasting

1. Proper recording of data.
2. Careful **study of synoptic charts**.
3. Search for similar situation from the historical data.
4. Preparation of the weather **condition charts in 24 hours**.
5. Drawing **quick, correct and definite** conclusion.

Methods Of Forecasting

The techniques in the forecasting **are divided into three broad categories-**



Flow chart 3 -Methods of Forecasting

1. Synoptic Method:

A synoptic method involves detailed analysis of current weather **reports from a large area**.

- ✓ The current weather patterns **are related to the past and analogous situations** and **forecast are prepared on the assumption** that a current situation will behave on the lines of the past.
- ✓ Useful for **short range forecasts**.

2. Statistical Method:

Regression equations or other **advanced relationships are formed between various weather elements** and the subsequent climate.

- ✓ Predictions is **based on physical interaction** or relationship with the predicant.
- ✓ Here to predict a **rainfall IMD uses 16 parameters**.
- ✓ Suitable for **long as well as short range forecast**.

3. Numerical Weather Prediction Techniques:

Numerical weather prediction definition states that it forecasts **weather using statistical models of the atmosphere** and oceans dependent on current weather conditions.

- ✓ The action of the atmosphere is expressed in this system by a series of equations based on physical **laws governing airflow, air pressure, and other data**.
- ✓ The method is used for **medium-term forecasts**.

Types of weather forecasting-based on time period

1. Now casting –

A weather forecast in which the details about the current weather and forecasts up to **a few hours ahead (but less than 24 hours)** are given.

2. Short range weather forecasting–

Short range weather forecasts are weather forecasts valid up to **72 hours ahead**.

- ✓ This **forecast range is mainly concerned with the weather systems** observed in the latest weather charts and also by considering the generation of new systems within the time period.

3. Medium range forecasts-

These are weather forecasts generally valid for a period **of 4 to 10 days (However, up to 7 days in tropics)**.

- ✓ In this, the **average weather conditions and the weather on each day will be prescribed** with progressively lesser details and accuracy than that of the short-range forecasts.

4. Long range forecast-

Long range forecast is defined as the **forecast from 30 days' up to one** season's description of averaged weather parameters.

- ✓ The seasonal forecast comes **under long range** forecast.

5. Extended range forecast-

This is the forecast range lies between long range (seasonal) and medium range.

- ✓ it starts generally from days **10 (however, beyond days 7 in tropics)** up to one month.

Table 2 -Time Period of Weather Forecasting

S. No	Range Of Weather Forecast	Time Period
1.	Now casting	Few hours ahead but less than 24 hours
2.	Short range	Up to 72 hours
3.	Medium range	4 to 10 days
4.	Extended range	From 10 days to month or till one season
5.	Long range	30 days to a season

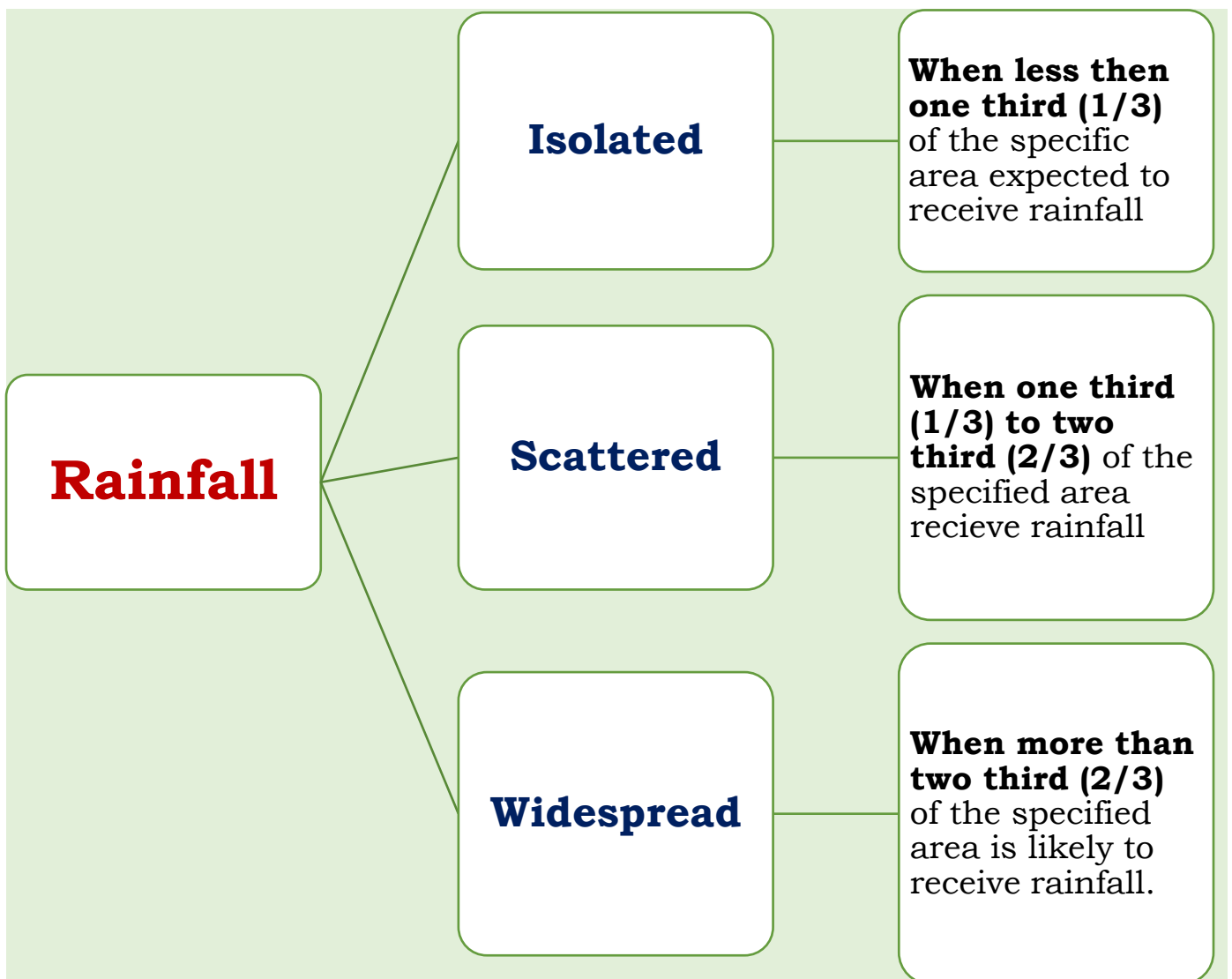
Forecasting information

1. **Sunshine**- the information of sunshine is provided in the forecast so as to help the farmers in taking decision on operations like harvesting crops, drying the crops etc.

Sunshine- available sunshine per day expressed as a cloudiness on an average for the entire day.

- ✓ **Sunny day** – when sky is clear with **90 to 100 %** sunshine
 - ✓ **Partly cloudy or scattered cloudiness**-when **50 to 90 % sunlight** on a particular day.
 - ✓ **Considerable cloudiness**- for **20 to 50 % sunshine**.
 - ✓ **Mostly cloudy**- 20% or **less than 20%** sunshine.
2. **Precipitation/ Rainfall** – a forecast of rainfall is the most important aspect of weather for Indian agriculture

Flow Chart 4- Rainfall Forecasting



3.Dew-

Dew forecast in 6 categories that is **very light, light, moderate, heavy, very heavy and wet.**

COLOUR CODE FOR WEATHER WARNINGS

The following colour codes are used in weather warnings for bringing out the severity of the weather phenomena expected.

WARNING (TAKE ACTION)
ALERT (BE PREPARED)
WATCH (BE UPDATED)
NO WARNING (NO ACTION)

Image (iii)-warning signs for extreme weather

CARDINAL TEMPERATURE

It is the **minimum, optimum and maximum temperature required** for the crop growth. Every crop has different cardinal temperature.

Table 3- Cardinal Temperature for Crops

Crops	Minimum	Optimum	Maximum
Oat	4 - 5	25	30
Maize	8 - 10	32 - 35	40 - 44
sorghum	8 - 10	32 - 35	40
Pearl millet	8 - 10	30 - 32	40
Berseem	3 - 4.0	25 - 27	32
Wheat	3 - 4.5	25	30
Rice	10 - 12	30 - 32	36 - 38

